

Aspect-Based Sentiment Analysis for Vietnamese Feedback about Sneakers on Shopee

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Abstract

Nowadays, shopping on e-commerce platforms has become more and more popular and the review will help a lot for customers to make decisions. Much research has used machine learning to classify the sentiment of the review but traditional sentiment analysis can not detect the aspect of the product or service of the seller, they can not optimize and exploit the full value contained in the data. Lately, sentiment analysis has been upgraded to a new age with aspect base analysis that can detect the sentiment for each aspect of the product. In this paper, we present a handmade dataset with 11700 reviews on shopee. We also implement many Deep Learning models with two approaches: Single-Task Learning and Multi-Task Learning for this dataset to find out the efficiency of the dataset and models for two tasks: aspect detection and sentiment classification.

1 Introduction

When customers want to buy something on e-commerce platforms they will reference the review from the person who has bought the product before. With the explosion of e-commerce more and more people join the e-commerce platforms and the data becomes huge. This data combines a lot of valuable resources for the customer and also the seller. With the customer the data will help them make a decision whether to buy or not. With the seller the data helps them improve the aspect that they are not well. It is a motivation to create a model to detect the aspect of e-commerce data.

Sentiment analysis (SA) is a process that classifies whether a data is positive, negative or neutral to help the user to know more insight about the data. In this paper with the customer, the sentiment analysis helps them to know more about the product, know more about the quality, the outlook from the buyer before what they should choose, and help them to make the decision. With the seller, the shop sentiment analysis helps them know better about

themselves if they were good enough or not, and helps them to improve themselves. But traditional sentiment analysis can not analyze the details of the specific aspects, components, parts, or functionalities of the entity. To do that, some researchers have presented a new model called aspect-based sentiment analysis that can analyze the details of the entity. Aspect-based Sentiment Analysis is a variety of sentiment analysis that helps in the improvement of the business by knowing the features in their product which they need to improve according to customer's feedback to make their product a best seller. ABSA identifies the aspects in the given review about a product and also finds if the aspect mentioned in the review belongs to which class of sentiment.

Lately, Aspect-based Sentiment Analysis has attracted a lot of attention with its technical and approaches. The main task of Aspect-based Sentiment Analysis is defining the aspect of the entity, extraction of the linguistic expression, and classifying the opinion or sentiment for each aspect.

In this paper:

Input: A customer feedback A from one of the popular E-commerce platforms is Shopee about a sneaker.

Output: one or many customer opinions that are extracted from feedback A.

Nowaday, in Vietnam e-commerce platforms have appeared more but most of them have a feedback system that rate by point or star, rare of them has sentiment analysis and almost none of them has Aspect-based Sentiment Analysis. We believe that with Aspect-based Sentiment Analysis e-commerce platforms will attract more users with Aspect-based Sentiment Analysis convenient. If an e-commerce platform applies Aspect-based Sentiment Analysis it can focus on the aspect-based that the customer cares about. Mining and summarizing opinions from text about specific entities and their aspects can help consumers decide what to purchase and

Aspect	Definition
PRICE	Reviews mention price of the sneakers, whether it is cheap or expensive
SHIPPING	The users comment about the delivery time and shipper's attitude
OUTLOOK	Reviews describe the style and form of sneakers
QUALITY	Review the quality of the product, including the comfort, fabric, and durability
SIZE	Comments mention the size of the sneakers, whether it fit or not
SHOP SERVICES	The reviews refer to customer care service and packing
GENERAL	Reviews describe the sneakers in general terms without any specific aspects
OTHERS	Spam reviews

Table 1: Aspect definition

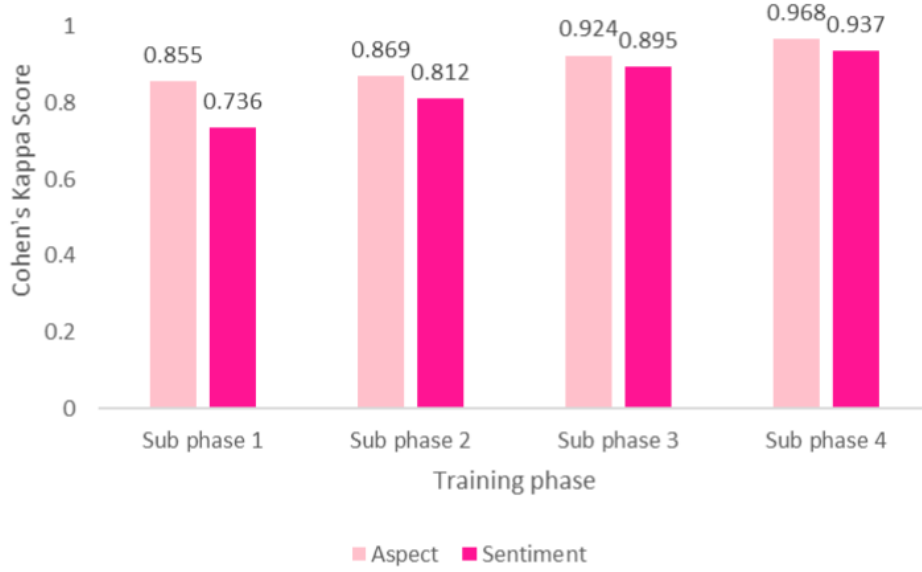


Figure 1: Cohen's Kappa Score of four training sub phases

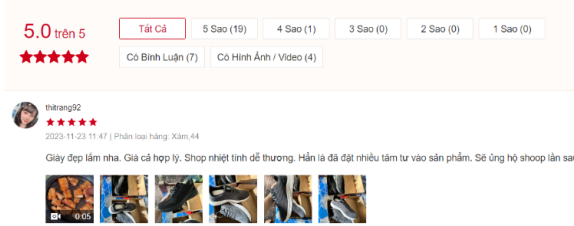


Figure 2: A feedback from Shopee

businesses to better monitor their reputation and understand the needs of the market

To do that, this paper presents a dataset with 11700 feedback comments about sneakers on shopee. 7 aspects with all annotated manually

2 Related Work

ABSA was introduced as a SemEval task in 2014 (SE-ABSA14) providing benchmark datasets of English reviews and a common evaluation framework (Pontiki et al., 2014); the datasets were anno-

tated with aspect terms (e.g. “hard disk”, “pizza”) and their polarity for laptop and restaurant reviews, as well as coarser aspect categories (e.g., FOOD) and their polarity only for the restaurants domain. The task was repeated in SemEval 2015 (SE-ABSA15) aiming to facilitate more in-depth research by providing a new ABSA framework in which all the identified constituents of the expressed opinions (aspects, opinion target expressions and sentiment polarities) meet a set of guidelines/specifications and are linked to each other within tuples. In the context of the new framework an aspect category is defined as a combination of an entity type E (e.g. LAPTOP, KEYBOARD, CUSTOMER SUPPORT, RESTAURANT, FOOD) and an attribute type A (e.g. USABILITY, QUALITY, PRICE) of E, making more explicit the difference between entities and the particular facets that are being evaluated (Pontiki et al., 2015). The SemEval-2016 task-5 (SE-ABSA16) (Pontiki et al., 2016) dataset extended SE-ABSA15 to new do-

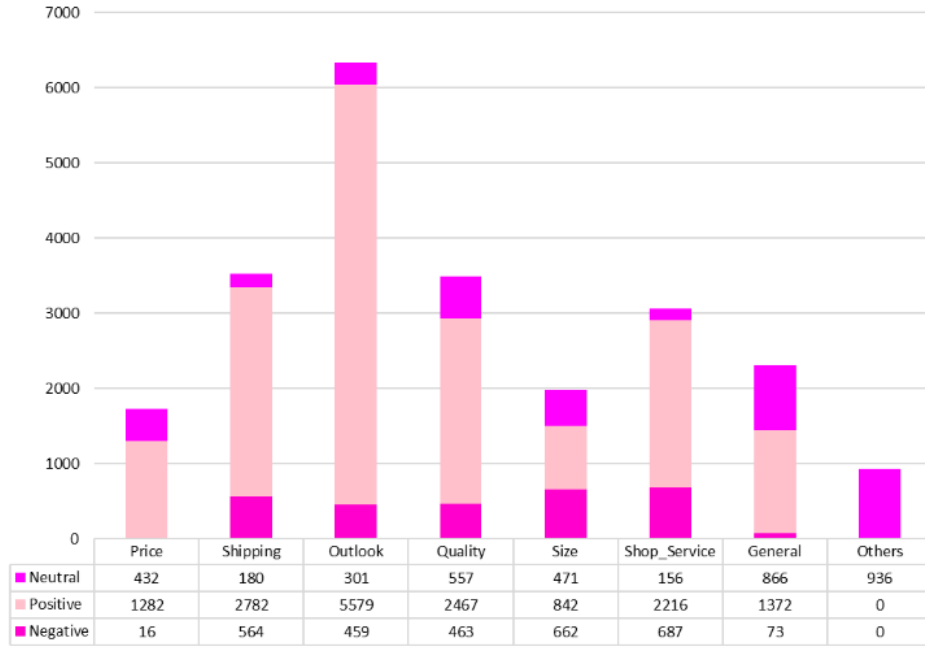


Figure 3: Aspect and Sentiment distribution in the dataset

mains such as Hotels, Consumer Electronics, Telecom, Museums, and other languages. In several recent years, many researchers have published their works about ABSA with positive results ((Nazir et al., 2020), (Do et al., 2019)).

In Vietnam many universities and researchers have published their datasets, works in ABSA: Vietnamese ABSA corpus about smartphone reviews (Mai and Le, 2018), SA-VLSP2018 dataset (published by H. T. Nguyen et al.) about hotel and restaurant reviews. A dataset on the same domain as VLSP was created by (Nguyen et al., 2019) for the two tasks of sentiment classification and aspect extraction that were the focus of the earlier work. In addition, (Thin et al., 2021) constructed a sentence-level dataset for the same topic that was annotated with high inter-annotator agreements in two earlier researches. UIT-ViSFD benchmark dataset is created for evaluating ABSA for mobile e-commercial by (Luc Phan et al., 2021). (Thanh et al., 2021) presented UIT-ViSD4SA dataset with span detection for ABSA, which is a benchmark Vietnamese smartphone feedback dataset. And most recently, (Tran et al., 2022) performed ABSA on a dataset about e-commercial beauty product reviews.

3 Data

3.1 Sources and Crawling Method

Our dataset was crawled on Shopee about sneakers for aspect-based sentiment analysis tasks by

using Selenium in Python. We choose Shopee because it is a popular e-commerce platform in Vietnam, it has more feedback than other platforms. This dataset contains 11700 Vietnamese reviews and was annotated manually for 2 subtasks: aspect detection and sentiment classification. With our dataset, we design 8 aspects: PRICE, SHIPPING, OUTLOOK, QUALITY, SIZE, SHOP-SERVICES and GENERAL and 1 aspect to detect spam OTHERS, and their definition are shown in Table 1. For sentiment classification, we divided it into 3 labels: POSITIVE, NEGATIVE and NEUTRAL. We split data into 3 pieces in the ratio 7:2:1: training set with 8424 reviews, testing set with 2340 reviews, and validation set with 936 reviews.

3.2 Annotation Process

First of all, we define a basic annotation guideline. For training, we randomly take 100 reviews to annotate, then we calculate Cohen’s Kappa score for those annotated reviews. For each aspect or sentiment that has not enough 4 agreements, we gather and discuss a new rule to update our guideline. We trained four rounds to obtain Cohen’s Kappa score higher than 90% before performing data annotation independently. After the training phase, the rest of the data was split into 4 parts and each member annotated 1 part following the rules in our annotation guideline.

Set	Reviews	Avg aspect/review	Positive	Negative	Neutral	Total sentiment
Train	8424	1.99	11921	2107	2776	16804
Val	936	2.03	1346	221	334	1901
Test	2340	1.99	3273	596	789	4658

Table 2: The overview statistics of the dataset

Reviews	Aspect	Polarity
Mình đặt size 39 nhưng chật k đeo nổi. Còn giày thì đẹp giao hàng nhanh (I ordered size 39 but it's too tight to wear. The sneakers are beautiful and the shipping is fast)	SIZE OUTLOOK SHIPPING	NEGATIVE POSITIVE POSITIVE
Hình ảnh chỉ mang tính chất nhận xu (Image is only for receiving coins)	OTHERS	NEUTRAL
Giày đẹp nhưng đóng gói sơ sài nên vận chuyển xa làm giày bị biến dạng (The sneakers is beautiful but poorly packaged so long distance transportation made it become deformed)	OUTLOOK SHOP_SERVICE	POSITIVE NEGATIVE

Table 3: Some examples of the dataset

3.3 Statistics

Our dataset contains 11,700 reviews about sneakers with 23,363 pairs of aspects and sentiments. The overview statistics of training, validation and testing set are shown in Table 2. Notably, the OUTLOOK aspect has the highest number of reviews, followed by SHIPPING and QUALITY. In terms of sentiment distribution, POSITIVE labels dominate across all aspects (accounting for from 43% to 88%), aligning with the common trend of customers expressing satisfaction more frequently than dissatisfaction. The PRICE aspect stands out with a remarkably low count of NEGATIVE feedback suggesting overall POSITIVE or NEUTRAL sentiments regarding pricing. However, there is an observable imbalance in label counts within each aspect, which may impact the learning of the models. With separate emotional distribution in aspects, the SIZE aspect exhibits a notable number of NEGATIVE feedback, warranting a closer investigation into potential areas for product improvement. Conversely, most other aspects boasts a significant number of POSITIVE feedback, indicating a positive customer service experience. Overall, this dataset provides a valuable foundation for sentiment analysis, with further exploration recommended to uncover deeper insights into specific aspects and sentiments.

3.4 Data Preprocessing and Word Embedding

For data preprocessing, we have applied many techniques:

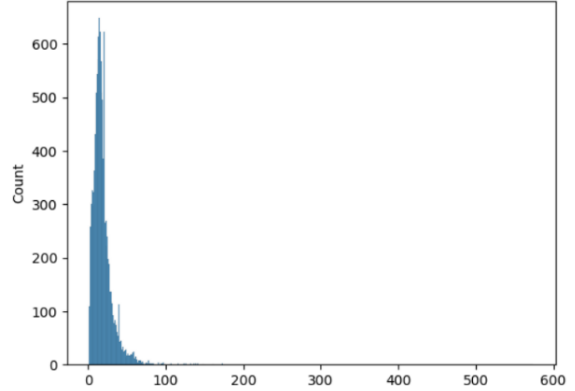


Figure 4: Word length distribution of the dataset

- Remove HTML
- Convert unicode, normalize acronyms, teen-code, emoji
- Remove unnecessary, duplication characters
- Tokenize reviews

For embedding, we use AIVIVN SentiTask pre-trained word embedding of ETNLP. Padding is used with a maximum length of 200, which is chosen based on the word length distribution chart of the dataset shown in figure 4.

4 Model

4.1 Approach

For Aspect Based Sentiment Analysis we have 2 ways to approach a single task and multitask.

Single task or Single-task learning is the process of learning to predict a single outcome (binary, multi-class, or continuous). According to (Zhang et al., 2022), there are 2 main subtasks in Aspect Based Sentiment Analysis: aspect category detection and aspect sentiment classification. Aspect category detection will detect the aspect and the output will be taken to a sentiment classification model where it will classify the sentiment of the review. To do this, we built 2 different models for each task,

By contrast, **Multi-task learning** is the process of jointly learning to predict multiple outcomes on inputs of the same modality. For the Aspect Based Sentiment Analysis, the aspects and corresponding sentiment are predicted simultaneously. With multi-task, our approach is a deep learning model with the ability to multitask.

We have referenced many papers with Aspect based sentiment analysis and finally we have made a decision. With a simple model we use **BiLSTM**, **BiGRU**, **BiLSTM/BiGRU + Conv1D**, **BiLSTM + BiGRU + Conv1D**. And for more complex, we choose two pre-trained models, **PhoBERT-base** and **ViSoBERT**.

4.2 BiLSTM

Bidirectional LSTM, or BiLSTM, is a sequence processing model that consists of two LSTMs: one taking the input in a forward direction, and the other in a backwards direction. BiLSTMs effectively increase the amount of information available to the network, improving the context available to the algorithm (e.g. knowing what words immediately follow and precede a word in a sentence).

4.3 BiGRU

A Bidirectional GRU, or BiGRU, is a sequence processing model that consists of two GRUs. one taking the input in a forward direction, and the other in a backwards direction. It is a bidirectional recurrent neural network with only the input and forget gates.

4.4 Conv1D

Conv1D is a layer that creates a convolution kernel that is convolved with the layer input over a single spatial (or temporal) dimension to produce a tensor of outputs.

4.5 PhoBERT-base

PhoBERT-base is a pre-trained language model that was made for Vietnamese. PhoBERT-base pre-training approach is based on RoBERTa which optimizes the BERT pre-training procedure for more robust performance. PhoBERT-base outperforms previous monolingual and multilingual approaches, obtaining new state-of-the-art performances on four downstream Vietnamese NLP tasks of Part-of-speech tagging, Dependency parsing, Named-entity recognition and Natural language inference.

4.6 ViSoBERT

ViSoBERT is the state-of-the-art language model for Vietnamese social media tasks: ViSoBERT is the first monolingual MLM (XLM-R architecture) built specifically for Vietnamese social media texts. ViSoBERT outperforms previous monolingual, multilingual, and multilingual social media approaches, obtaining new state-of-the-art performances on four downstream Vietnamese social media tasks.

5 Experiment

5.1 Experimental settings

With our data we deploy 7 different models to find out the best model for our dataset.

The number of units in BiLSTM or BiGRU is 400 units and the activation function in these layers is tanh function. In the Conv1D layer, the kernel size is 2 and the filter is 128, which means reducing the input dimension from 400 to 128. For ViSoBERT, we didn't preprocess reviews due to its feature.

We use binary_crossentropy loss function for the aspect detection task and categorical_crossentropy for the sentiment classification tasks. Adam and official.npl are used as Optimizer. We use batch size 128 and 70 epochs.

5.2 Evaluation Metrics

F1-score macro, which is the harmonic mean of the precision and recall, is used because of the imbalance of aspect and sentiment in the dataset. It thus symmetrically represents both precision and recall in one metric and give us a more general view of the models' performance. The formula of F1-score is as follow with TP, FP, and FN denoted for True Positive, False Positive and False Negative respectively:

	Aspect Detection		Polarity Detection		Aspect-polarity	
	STL	MTL	STL	MTL	STL	MTL
BiLSTM	0.8251	0.6481	0.7413	0.6479	0.6097	0.5671
BiGRU	0.8444	0.7725	0.7375	0.6751	0.6057	0.5936
BiLSTM + Conv1D	0.8421	0.7223	0.7592	0.6780	0.6306	0.5934
BiGRU + Conv1D	0.8504	0.7422	0.7732	0.6811	0.6468	0.5997
BiLSTM + BiGRU + Conv1D	0.8444	0.7479	0.7632	0.6915	0.6380	0.6077
PhoBERT-base	-	0.8461	-	0.7382	-	0.6519
VisoBERT	-	0.8449	-	0.7317	-	0.6314

Table 4: The overall experimental results

Aspect	Negative	Positive	Neutral
PRICE	0.00	0.9162	0.7895
SHIPPING	0.8872	0.9720	0.00
OUTLOOK	0.7716	0.9673	0.3333
QUALITY	0.7204	0.9215	0.5415
SIZE	0.7159	0.9277	0.5497
SHOP SERVICE	0.8153	0.9530	0.2000
GENERAL	0.00	0.8543	0.7471

Table 5: Sentiment Classification results of each aspect of BiGRU + Conv1D

Aspect	STL	MTL
PRICE	0.8403	0.8059
SHIPPING	0.9440	0.9534
OUTLOOK	0.9297	0.9265
QUALITY	0.8176	0.8191
SIZE	0.8600	0.8265
SHOPSERVICE	0.8559	0.8486
GENERAL	0.6526	0.6883
OTHERS	0.7978	0.7955

Table 7: Aspect Detection results of BiGRU + Conv1D and PhoBERT-base

Aspect	Negative	Positive	Neutral
PRICE	0.00	0.7985	0.4296
SHIPPING	0.7383	0.9370	0.1728
OUTLOOK	0.6420	0.9154	0.3894
QUALITY	0.4521	0.8275	0.3941
SIZE	0.5965	0.7524	0.5191
SHOP SERVICE	0.6894	0.8167	0.2667
GENERAL	0.1429	0.6106	0.5593

Table 6: Sentiment Classification results of each aspect of PhoBERT-base

$$\begin{aligned}
Precision &= \frac{TP}{TP+FP} \\
Recall &= \frac{TP}{TP+FN} \\
F1 &= \frac{2*Precision*Recall}{Precision+Recall} = \frac{2*TP}{2*TP+FP+FN}
\end{aligned}$$

5.3 Results and Discussion

Our model was successful in identifying and extracting important aspects from the input text. We have divided the results into 4 tables with different evaluation aspects using the F1-score index.

In the overall experimental results shown in Table 4 for the Aspect-Based Sentiment Analysis (ABSA) problem, our observations are directed towards two main tasks: single-task and multi-

task, and determine the performance of seven other models. In the single-task context, except for two pre-trained models that do not apply to this task. At Aspect Detection section, all deep learning models demonstrate high performance, achieving an average F1-score of nearly 0.84. Notably, the BiGRU model combined with the Conv1D layer achieved the highest F1-score with 0.8504. Through multi-task evaluation, pre-trained models PhoBert-base and Visobert stand out with significant performance. PhoBert-base achieved the highest F1-score of 0.8461, while Visobert ranked next with an F1-score of 0.8449. In contrast, the remaining five deep learning models only achieved poor performance with an average score of only 0.72. In the polarity detection section, the BiGRU + Conv1D model and the pre-trained PhoBERT-base model continue to demonstrate the highest performance in both single-task and multi-task scenarios. For the BiGRU + Conv1D in the single-task setting, this model achieved the highest F1-score of 0.7732, outperforming all other models with an average score of 0.75. The effectiveness of this model can be further analyzed in terms of how it handles and identifies aspects related to polarity in the text. PhoBERT-base, as a pre-trained model, main-

tains its strength in the multi-task environment, achieving the highest F1-score of 0.7382 in the Polarity Detection task. This stands out compared to the poorer performance of the remaining six models, which only achieved an average score of 0.68. Detailed analysis may reveal how PhoBERT-base integrates information from both tasks to enhance performance. In the Aspect-Polarity Detection section, considering the results from the previous two sections, it comes as no surprise that both the BiGRU + Conv1D and PhoBERT-base models continue to exhibit the best performance. For BiGRU + Conv1D, the F1-score is 0.6468, surpassing the average score of the other models, which is 0.6210. This highlights the consistency of BiGRU + Conv1D in maintaining high performance across different aspects of aspect-polarity detection. Similarly, PhoBERT-base achieves the highest performance with an F1-score of 0.6519, outperforming the other models with an average score of 0.5988. This reaffirms the effectiveness of PhoBERT-base in capturing and analyzing aspect-specific sentiment. In summary, the dominance of BiGRU + Conv1D and PhoBERT-base across all three sections underscores their robustness and versatility in addressing the complexities of Aspect-Based Sentiment Analysis. These results contribute to a comprehensive understanding of the strengths and capabilities of these models in handling various facets of ABSA tasks.

From the results we obtained from the previous table, we use the 2 models that have achieved the highest performance for all 3 approaches: BiGRU + Conv1D and PhoBERT-base models to continue evaluating aspects. From Table 7, we can see that the BiGRU + Conv1D model demonstrated superiority in assessing sentiments related to PRICE (0.8403 vs 0.8059), SIZE (0.8600 vs 0.8265), and SHOP SERVICE (0.8559 vs 0.8486), while PhoBERT-base outperformed in aspects such as GENERAL (0.6883 vs 0.6526), SHIPPING (0.9534 vs 0.9440). Both models exhibited robust performance in evaluating sentiments about QUALITY, with PhoBERT-base holding a slight edge.

As you can see in Table 5, certain aspects such as PRICE with a NEGATIVE label, SHIPPING with a NEUTRAL label, and GENERAL with a NEGATIVE label were not detected by the BiGRU + Conv1D model. This can be explained by a severe imbalance in the data, particularly in these

aspects, as clearly demonstrated in the Aspect and Sentiment distribution in Figure 3. This imbalance may lead to misunderstandings and shortcomings in the training process of the model for these specific aspects. Nevertheless, the BiGRU + Conv1D model still demonstrates impressive performance with other aspects, with F1-scores ranging from 0.7159 to 0.9530. However, there are three specific cases, namely QUALITY, SIZE, and SHOP SERVICE with a NEUTRAL label, where the model did not achieve satisfactory results. Specifically, the scores for these aspects are 0.5415, 0.5497, and 0.2000, respectively, indicating challenges in handling reviews and emotions for these aspects within the context of the utilized model.

In a structure similar to Table 5, Table 6 provides insights into the sentiment classification results for each aspect using the PhoBERT-base model in the context of Aspect-Based Sentiment Analysis (ABSA). Diverging from the BiGRU + Conv1D model, PhoBERT-base achieved the classification of GENERAL with a NEGATIVE label and SHIPPING with a NEUTRAL label, albeit with relatively low F1-scores of 0.1429 and 0.1728, respectively. Notably, both models share the limitation of failing to classify the PRICE aspect with a NEGATIVE label. While the PhoBERT-base model exhibits improvement in classifying labels with limited data, it appears that its overall performance is not as high as that of the BiGRU + Conv1D model. Specifically, numerous aspects fall below the threshold of a score of 7, with the most challenging aspects being SHOP SERVICE, QUALITY and OUTLOOK with a NEUTRAL label, achieving scores below 4. This suggests a notable weakness in the PhoBERT-base model when handling sentiment analysis for these specific aspects, emphasizing the importance of further refinement or exploration of alternative models for enhanced performance in diverse ABSA scenarios.

5.4 Error analysis

We have found that all models have three types of errors that misclassify the sentiment polarity, cannot detect aspects and detect wrong aspects. As you can see in the first comment of table 8, which is several predictions of PhoBERT-base model, “shop thái độ lồi lõm quá” is supposed to be SHOP_SERVICE#NEGATIVE but the model classified it as SHOP_SERVICE#POSITIVE. In the second review, the model cannot detect SPAM

Reviews	Aspects	Sentiments
Giao hàng nhanh nhưng shop thái độ lỗi lổm quá	SHIPPING SHOP-SERVICE	POSITIVE POSITIVE
Áo ma lazada quá	NONE	NONE
Hàng đóng gói rất đẹp. Quả nhiên người bán đã gửi gắm rất nhiều tâm tư và tình cảm trong lúc gói hàng. Hộp không có dấu hiệu bị móp, chứng tỏ bộ phận vận chuyển đã nâng như nâng trứng húng như húng hoa. xin nghiêng mình kính cần cảm ơn vì tất cả	SHIPPING OUTLOOK SHOP_SERVICE	POSITIVE POSITIVE POSITIVE

Table 8: Some incorrect prediction examples of PhoBERT

aspect due to the fact that most SPAM reviews in the dataset are of the form “Hình ảnh chỉ mang tính chất nhận xu” or advertisement so this is a sentence that the model may not have learned yet. For the final example, it only contains two aspects: SHIPPING and SHOP_SERVICE, but the model predicted 1 more wrong aspect, OUTLOOK.

6 Conclusion and Future work

In this paper, we have presented our approach to the aspect-based sentiment analysis of sneaker products on Shopee. We have presented our new dataset with 11700 reviews on Shopee about sneakers for aspect extraction and sentiment analysis tasks. We also deployed many models for our data and the results show that BiGRU + Conv1D and PhoBERT-base have the highest F1-score for single-task and multi-task learning, respectively.

In the future, we will research triplet extraction and multimodal aspect-based sentiment analysis in order to build a system that has the ability to automatically collect, process, analyze the feedback and visualize the result. In our opinion, this system will be a big upgrade for many e-commerce platforms. This system will not only help customers find their best suitable product easier but also help sellers enhance their products’ aspects that are not good enough. Furthermore, we will develop this system to adopt for another domains and Social Listening.

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